

B. Tech 3rd Semester 1st Sessional Examination 2025
Sub: Mathematics for Civil Engineering (BMA23211T)
Branch- CE (A) , Total Marks-30, Date: 10/09/2025

1. Find the Laplace Transform of (any three)

5X3=15

(a) $\sin 6t \sin 4t$

(b) $\cos^3 2t$

(c) $t^3 e^{2t}$

(d) $t^2 \sin t$

(e) $e^{-3t} (2 \cos 5t - 3 \sin 5t)$

(f) $\frac{e^{at} - e^{bt}}{t}$

2. Evaluate (any three):

5X3=15

(a) $L^{-1} \left\{ \frac{1}{s^2 - 3s + 2} \right\}$

(b) $L^{-1} \left\{ \frac{3s+1}{(s-1)(s^2+1)} \right\}$

(c) $L^{-1} \left\{ \frac{s}{s^2 + 8s + 16} \right\}$

(d) $L^{-1} \left\{ \frac{s+2}{s^2 - 4s + 20} \right\}$

(e) Evaluate: $L^{-1} \left\{ \frac{1}{(s+1)(s+2)} \right\}$ using Convolution theorem

(f) Evaluate: $L^{-1} \left\{ \frac{1}{s(s^2+1)} \right\}$ using Convolution theorem